ToolBook 8 Reverse Engineering

Overview & Status Report

October 2026 — Nida ATT Master_3 Courses

1. What These Books Are

The course files are classic Asymetrix/Click2learn ToolBook books (ToolBook II 8.1 shells, book version 01 00 00 08). Each **.tbk** begins with a short PE stub (0xC00 bytes, no real code) followed by the book body. The magic **ZZZTBKBOOKLOC** and the value **0x22222222** mark the start of the real book data. The opener is **tbload.exe** via **MINSTB32.dll**.

The runtime engine that actually interprets the book is the 16-bit NE binary **TB80RTM.EXE** (unpacked from its SLR Optloader wrapper). Supporting DLLs (**TB80BAS**, **TB80CMP**, **TB80BMP**, **TB80UTL**, **TB80W16**, and others) supply object types, painting, OpenScript compilation/expansion, bitmaps, and windowing.

The courses themselves are Navy electronics CAI. **courses.xml** is a menu export that lists 0_MASTER3 plus twenty rating tracks. On disk there are hundreds of lesson **.tbk** files. Work has concentrated on **9e954d01.tbk** (Logic Families) plus the shared shell **nida81.tbk** and the system books.

2. On-Disk Structure (Understood)

Page objects are the only records that begin with the marker **80 65 00 05 00 02 02**. Inside each page the widgets are nested chunks that begin **14 00 00**. A typical property block starts **14 00 00 01**. Field text is stored as a Text property (length-prefixed, often with a leading CRLF).

Standard property IDs (bounds = 0x4004, text = 0x402e, script = 0x4023, vertices = 0x4034, etc.) were recovered from NE resource type 0x2d1 / id 0x2d0 in TB80BAS.

Object-type names (Page, Button, Field, Picture, Polygon, Arc, Curve, Rectangle, etc.) live in a classification table inside TB80UTL.DLL and are also present as resources in the system .sbk files.

3. What Modern Readers Already Do

Several generations of clean-room readers exist:

• Early HTML/XML viewers (viewer, viewer2, viewer3) that walk the page marker, extract text and quoted jumps.
• Maybe2 / Maybe4 lessonshells: JSON with pages, Next/Back, quiz choices, and decoded script.
• runtime/ and runtime_cpp/: Python and C++ packages that parse a book and emit HTML or drive a simple player.
• Brickwall / Foolbook: self-contained GUI readers that show page lists, field text, and quoted jumps.

All of these can already display the textual content, navigation, and quizzes of books that resemble Logic Families. They do not yet paint arbitrary polygons, arcs, curves, or placed bitmaps for the full corpus.

4. Function Naming Status

Pattern-based naming across decompiled modules has identified ~1,918 nameable functions (~17.5% of analyzed). Manual confirmation has produced over 100 named functions, concentrated in paint, scale, property, memory, and string helpers.

See **Modules/** subdirectories for per-module confirmed names and nameable lists. See **03_Naming_Progress.pdf** for the full summary.

5. Open Work

• Bounds extractor (record-to-object+0xF) still missing from decompiles.
• Full OpenScript execution and scoring.
• Complete vector painting (arcs, curves, computed brackets) and bitmap placement.
• NE host that fully opens a book under emulation.

Document Map

This package is organized for printing:

Main directory — overarching documentation (this overview, architecture, naming progress).
Modules/tb80xxx_knownnames/ — confirmed function names and nameable lists per DLL.
BookFormat/ — on-disk record layouts, polygon/arc geometry, property IDs.
OpenScript/ — bytecode decoder notes and extracted handlers.
SoftICE/ — dynamic analysis notes and live object layouts.
Runtimes/ — clean-room reader status.